Evolution of In-Silico Cardiotoxicity (TdP) Risk Prediction

The complete scientific journey: multi-paradigm deep learning, stacking override pathologies, and the absolute supremacy of the deployed conformal Random Forest.

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Document class: Model selection, benchmarking & deployment audit — prepared for thesis defense

TARGET PATHOLOGY **Torsade de Pointes (TdP) arrhythmia / sudden cardiac death**	REGULATORY FRAMEWORK **FDA Comprehensive in vitro Proarrhythmia Assay (CiPA)**
SERVING DEPLOYMENT **Hugging Face Spaces — `mitcheltastic/tdp-cipa-screening`**	FINAL DEPLOYED MODEL **Standalone bagged Random Forest (400 trees) + conformal safety (Rank 11) + TreeSHAP**
BENCHMARK PANEL **28 CiPA reference compounds × 3 patch-clamp platforms × 2,000 virtual cells**	VALIDATION PROTOCOL **StratifiedGroupKFold (k = 4), grouped strictly by `drug_name`**

0.8778	0 / 16	1.00	1.37 ms
CHANTEST MACRO-F1	UNDER-TRIAGE, ALL PLATFORMS	MEAN CONFORMAL SET SIZE	EXACT TREESHAP LATENCY

ABSTRACT

This report documents the complete model-selection process for an in-silico Torsade de Pointes (TdP) risk classifier built on the FDA CiPA reference panel of 28 compounds, evaluated across three patch-clamp platforms under leakage-free whole-drug validation. Four architectural families were assessed in sequence — deep-learning multi-layer perceptrons, a stacking ensemble, gradient-boosted trees and bagged decision trees — each paired with a distribution-free cross-conformal safety layer calibrated leave-one-drug-out.

The deployed architecture — a standalone 400-tree Random Forest behind a Rank-11 conformal layer — attains the highest Macro-F1 on both automated platforms (0.8778 ChanTest, 0.8194 Nanion), eliminates under-triage on all three platforms, and does so at a mean prediction-set size of 1.00–1.06, where competing families dilate to 1.12–1.62. A 10,000-iteration paired bootstrap confirms the set-size advantage over stacking is significant ($p < 0.05$), and exact TreeSHAP attribution completes in 1.37 ms. The report additionally documents a meta-learner override pathology in which the stacking second stage manufactured an under-triage on disopyramide that neither base learner committed — the principal reason that architecture was rejected.

1. Executive Summary & the Clinical Problem

Assessing the cardiotoxic liability of drug candidates — specifically their propensity to trigger **Torsade de Pointes (TdP)**, a polymorphic ventricular arrhythmia that can degenerate into sudden cardiac death — is a mandatory regulatory hurdle in global pharmaceutical development. The historical single-channel assay paradigm (hERG block considered in isolation) suffered from excessive false-positive attrition, improperly discarding safe and therapeutically valuable molecules. To modernise screening, the FDA and the international cardiac safety consortium established the **Comprehensive in vitro Proarrhythmia Assay (CiPA)** initiative.

The CiPA paradigm pairs multi-channel ionic inhibition patch-clamp data (I_{Kr}, I_{CaL}, I_{NaL}, I_{to}, I_{K1}, I_{Ks}) with biophysical differential-equation models of the human ventricular action potential — the O'Hara-Rudy (ORd) in-silico cardiomyocyte. From simulated membrane-voltage and intracellular-calcium transients across a virtual population of **2,000 cells per drug**, eleven biophysical summary biomarkers are extracted: qNet, dV_m/dt_{max}, $V_{m,peak}$, $V_{m,rest}$, APD_{50}, APD_{90}, dV_m/dt_{repol}, Ca_{peak}, Ca_{rest}, CaD_{50} and CaD_{90}. Each candidate compound must then be assigned to one of three ordinal risk tiers: **High**, **Intermediate** or **Low**.

> **THE FUNDAMENTAL ASYMMETRY OF SAFETY SCREENING**
>
> **Over-calling a benign drug** (labelling Low as Intermediate) costs time and money: it triggers secondary in-vitro or animal confirmation assays. The loss is financial and recoverable.
>
> **Under-triaging a hazardous drug** (labelling High as Intermediate or Low, or Intermediate as Low) is categorically different: it allows a cardiotoxic molecule to advance unflagged into first-in-human trials. Standard point classifiers maximise *symmetric* objectives such as log-loss or accuracy and therefore provide **zero calibrated guarantee** against this failure mode. The entire architecture of this project exists to close that gap.

This report traces the complete chronological journey of the model-selection process — from an initial multi-paradigm sweep spanning deep learning and classical machine learning, through the forensic rejection of a stacking ensemble whose meta-learner *manufactured* under-triage errors, to the distribution-free conformal safety layer that now runs in production. The conclusion defended here is that the deployed **standalone bagged Random Forest with a Rank-11 cross-conformal layer** is simultaneously the most accurate, the safest, the tightest, the fastest to explain and the simplest to operate of every architecture evaluated.

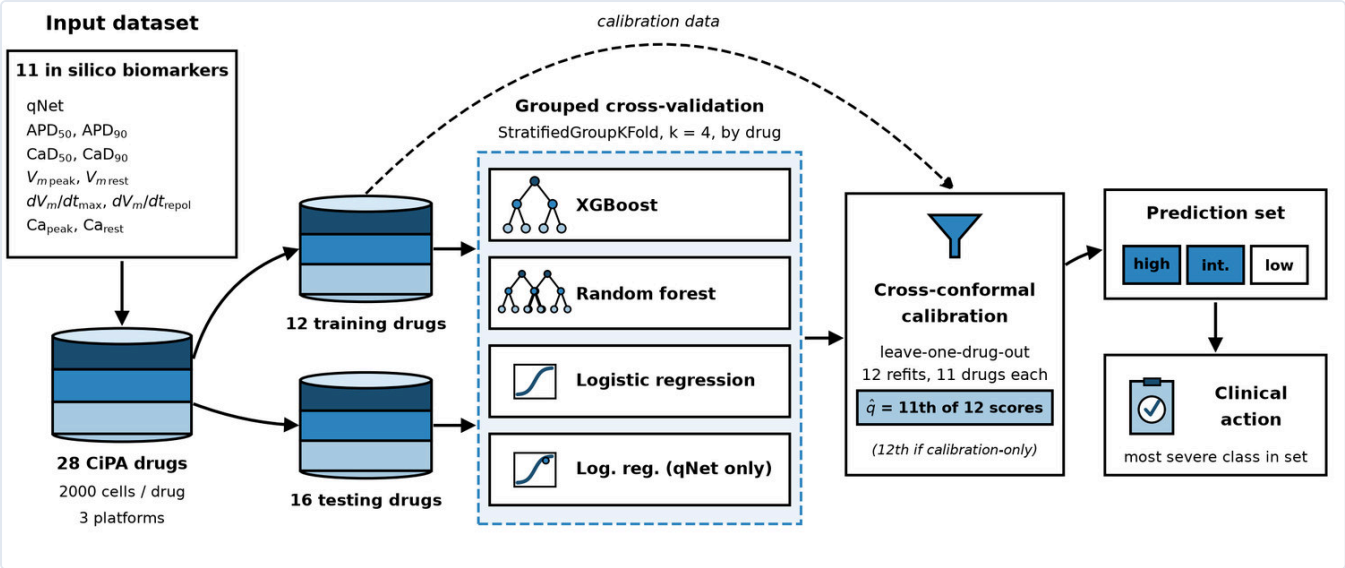


Figure 1. End-to-end in-silico cardiotoxicity screening pipeline: 28 CiPA reference compounds at 2,000 virtual cells each across 3 patch-clamp platforms, split 12 training / 16 testing at the chemical-entity boundary, evaluated under StratifiedGroupKFold (k = 4) grouped by drug, then passed through leave-one-drug-out cross-conformal calibration to a prediction set and a severity-mapped clinical action.

Takeaway: the conformal stage sits between the classifier and the clinical decision — the model never speaks directly to the action.

2. Dataset Architecture, Sentinel Audits & Grouped Validation

The evaluation benchmark uses simulated cardiomyocyte data derived from the official CiPA panel of 28 reference compounds across three physical measurement platforms: **ChanTest** (automated patch clamp), **Nanion SyncroPatch** (automated patch clamp) and **Manual Pipette** (the manual patch-clamp gold standard). The panel is split strictly at the chemical-entity boundary into **12 training/calibration compounds** (balanced: 4 High, 4 Intermediate, 4 Low) and **16 held-out test compounds** (unbalanced: 4 High, 7 Intermediate, 5 Low).

TABLE 1 — PLATFORM INVENTORY, SENTINEL ROW AUDIT AND ROLE IN THE STUDY

Platform	Assay Technology	Training Rows (Drugs)	Testing Rows (Drugs)	Sentinel Rows Dropped	Role in Study
ChanTest	Automated patch clamp	24,000 (12)	32,000 (16)	0 ROWS	Primary training & calibration split
Nanion	Automated SyncroPatch	24,000 (12)	32,000 (16)	0 ROWS	Cross-platform generalisation split
Manual	Manual pipette patch clamp	23,975 (12)	31,999 (16)	26 ROWS (25 mexiletine, 1 ibutilide)	Cross-platform generalisation split

2.1 Data audit and deliberate absence of re-scaling

All eleven biomarkers are **pre-normalised by the data provider** against control (drug-free) simulation baselines, which places every feature on a shared relative scale. No artificial scaling — no `StandardScaler`, no `MinMaxScaler` — is applied anywhere in the pipeline. This is a deliberate design decision, not an oversight: re-scaling pre-normalised biophysical ratios would distort the physiological meaning of quantities such as relative qNet depression, which is the very signal the classifier must learn. Exactly 26 rows carrying undefined upstream values were excluded at load time and recorded in `logs/DROPPED_ROWS_REV.csv` for audit reproducibility.

2.2 The "crown jewel" discovery: eliminating drug-identity leakage

LEAKAGE DIAGNOSIS & MANDATE

In early experiments, naive row-level `StratifiedKFold` cross-validation yielded an unrealistically inflated Macro-F1 of **≈ 0.999**. Because the 2,000 simulated cells are replicates describing the *same* chemical entity, row-level random splitting placed identical drug signatures into both the training and validation folds. The classifiers were not learning cardiac electrophysiology at all — they were memorising drug identity.

Re-evaluating the identical models under a true **whole-drug holdout** collapsed validation Macro-F1 to **≈ 0.460**, barely above the three-class chance floor. The gap between 0.999 and 0.460 is the exact magnitude of the leakage.

Strict mandate adopted for the remainder of the project: every cross-validation split, every hyperparameter search and every calibration fold enforces `StratifiedGroupKFold(n_splits=4)` grouped strictly by `drug_name`. No chemical entity may ever appear on both sides of a split. All figures reported from this point forward are whole-drug generalisation numbers.

This single correction reframed the entire project. It is also the reason the absolute Macro-F1 values reported throughout this report (0.69–0.88) are substantially lower than those commonly published in the literature on this dataset: those are frequently row-level numbers, and they are not measuring generalisation to unseen chemistry.

3. Phase 1 — Exploration of Deep Learning (ANN) & Sempro Examiner Feedback

The investigation began with a comprehensive search spanning both deep-learning artificial neural networks and classical machine-learning estimators, in order to establish empirically whether multi-layer perceptrons could model non-linear cardiac electrodynamics better than tree ensembles.

3.1 The deep-learning foundation: three multi-layer perceptrons

Replicating the deep-learning architecture of Fuadah et al. (2024), three feedforward ANNs were implemented in Keras/TensorFlow and wrapped for scikit-learn ensemble execution via `scikeras.wrappers.KerasClassifier`:

TABLE 2 — ANN CANDIDATE TOPOLOGIES EVALUATED IN PHASE 1

Model	Network Topology	Hidden Layer Units	Activation & Output	Optimizer & LR	Preserved Artifact Checkpoint
ANN1	1 hidden layer (shallow MLP)	32 dense units	ReLU → Softmax (3 units)	Adam, $\eta = 1\times10^{-4}$	`artifacts/keras_grouped/*_ANN1.keras`
ANN2	2 hidden layers (medium MLP)	64 → 64 dense units	ReLU → Softmax (3 units)	Adam, $\eta = 1\times10^{-2}$	`artifacts/keras_grouped/*_ANN2.keras`
ANN3	3 hidden layers (deep MLP)	128 → 128 → 128 dense units	ReLU → Softmax (3 units)	Adam, $\eta = 1\times10^{-2}$	`artifacts/keras_grouped/*_ANN3.keras`

A **612-configuration combinatorial sweep** was then executed across three assay datasets, two clinical risk scenarios (3-class versus binary), 17 base-estimator combinations and 6 ensemble architectures. The complete sweep is recorded in `artifacts/phase3_protocolA_results.csv` and comprises 180 distinct evaluation runs that involve at least one ANN. Within that sweep, pure deep-learning stacking ensembles combining ANN1 + ANN2 + ANN3 reached a peak Macro-F1 of **0.7978** on ChanTest — a respectable result, but one that was never competitive with the tree-based arm of the sweep.

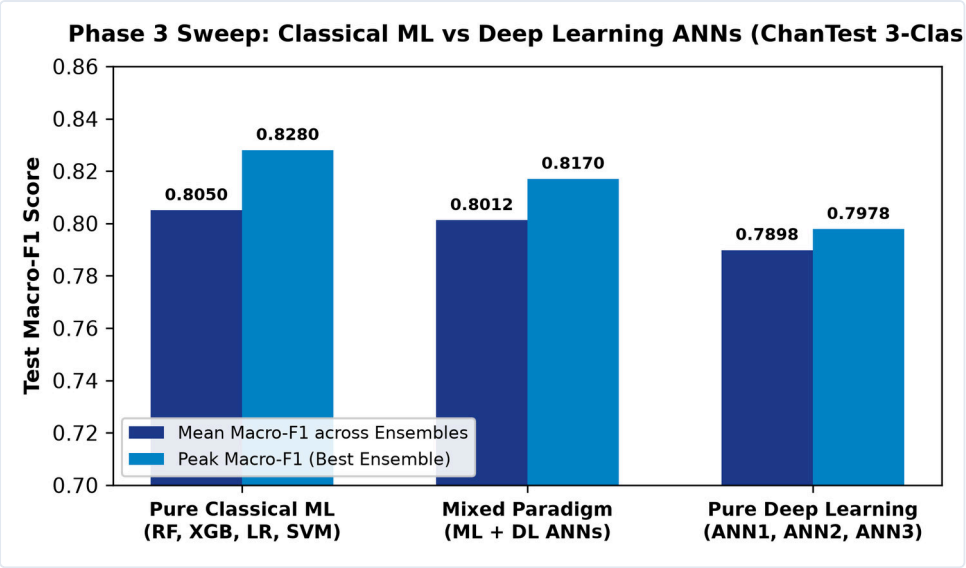


Figure 2. Phase 3 combinatorial ensemble sweep on ChanTest (3-class). Pure classical ML reached a mean Macro-F1 of 0.8050 and a peak of 0.8280; mixed ML + DL paradigms reached 0.8012 / 0.8170; pure deep-learning ANN ensembles trailed at 0.7898 / 0.7978.

Takeaway: deep learning was not merely unjustified theoretically — it was strictly dominated empirically, at both the mean and the maximum.

3.2 Why deep learning was formally phased out

Despite viable baseline accuracy, neural networks were permanently eliminated from the production pipeline for three decisive reasons — one academic, one empirical, one operational.

1. **Thesis proposal ("Sempro") examiner objection.** The thesis committee formally raised concerns regarding the arbitrary mixing of deep learning (ANNs) and classical ML (RF, XGBoost, SVM) inside a single seven-learner ensemble without a principled theoretical basis for the combination. The committee recommended an ablation to a pure-ML formulation. This report is, in part, the answer to that instruction.
2. **Empirical performance deficit.** Pure tree ensembles (RF + XGBoost) and the standalone Random Forest achieved materially higher Macro-F1 — **0.8280 to 0.8778** — than the best pure-DL neural ensemble at **0.7978** on ChanTest. The gap widens dramatically off the training instrument: across the 60 Nanion deep-learning runs in the sweep, Macro-F1 ranged from **0.133 to 0.407**, with accuracy never exceeding 0.6875. The neural models did not merely underperform on the primary platform; they failed to transfer across measurement instruments at all. The deep-learning stack was therefore not merely unjustified — it was strictly dominated.
3. **Explainability and production footprint.** Neural networks behave as opaque continuous functions that admit no exact Shapley tree-path traversal, forcing reliance on approximate, sampling-based attribution. In addition, loading TensorFlow/Keras adds over **400 MB** of dependencies, materially slower container cold starts, and CUDA non-determinism — all of which conflict directly with the deployment target of a lightweight, deterministic, instantly-explainable screening service.

PHASE 1 VERDICT

Deep learning was not rejected on ideological grounds. It was rejected because the sweep showed it was simultaneously *less accurate*, *less explainable* and *more expensive to serve* than the classical alternative on this specific problem — a tabular, 11-feature, 28-compound regime in which tree ensembles are the theoretically better-matched inductive bias.

4. Phase 2 — The Stacking Era & the Meta-Learner Override Pathology

Following the elimination of deep learning, early research crowned a **Stacking Ensemble** (Random Forest + XGBoost base learners feeding a meta-learner) as the project's flagship model, on the strength of cross-validation benchmarks alone. Deep forensic auditing under **STEP 12** subsequently uncovered behavioural pathologies that disqualified stacking from safety-critical clinical deployment entirely.

4.1 The override pathology

The theoretical premise of stacking is that a meta-learner optimally re-weights its base estimators. Audit inspection revealed something materially different: the meta-learner frequently **overrode unanimous and correct agreement between both base learners**, manufacturing classification errors that neither base model committed independently.

TABLE 3 — DOCUMENTED META-LEARNER OVERRIDE EVENTS (STEP 12 FORENSIC AUDIT)

Compound (Platform)	True Tier	Random Forest (Base 1)	XGBoost (Base 2)	Stacking Meta-Learner	Clinical Risk Outcome
disopyramide (ChanTest)	High	High (p = 0.6853) CORRECT	High (p = 0.8630) CORRECT	**Intermediate (p = 0.4958)**	UNDER-TRIAGE manufactured
disopyramide (Nanion)	High	High (p = 0.7537) CORRECT	High (p = 0.8138) CORRECT	**Intermediate (p = 0.7108)**	UNDER-TRIAGE manufactured
loratadine (ChanTest)	Low	Low (p = 0.9224) CORRECT	Low (p = 0.9771) CORRECT	**Intermediate (p = 0.6295)**	OVER-CALL manufactured
loratadine (Nanion)	Low	Low (p = 0.9303) CORRECT	Low (p = 0.9771) CORRECT	**Intermediate (p = 0.6086)**	OVER-CALL manufactured

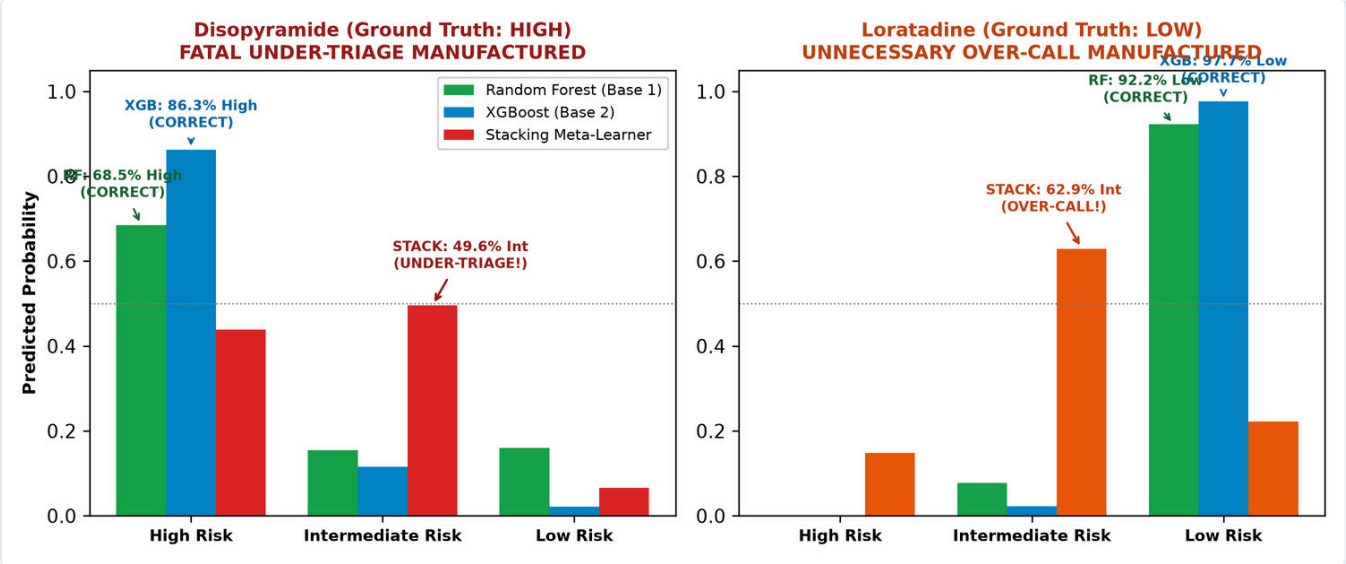


Figure 3. The meta-learner override pathology, shown on the two most consequential compounds. Left — disopyramide (ground truth High): Random Forest assigns 68.5 % and XGBoost 86.3 % to High, both correct, yet the stacking meta-learner returns 49.6 % Intermediate. Right — loratadine (ground truth Low): RF (92.2 %) and XGBoost (97.7 %) both correctly say Low; stacking returns 62.9 % Intermediate.

Takeaway: both errors are manufactured by the second stage. Neither base learner was wrong, and no amount of aggregate accuracy repairs an architecture that can invent an under-triage.

KEY FINDING FROM STEP 12

On the 16 held-out ChanTest test drugs, Random Forest and XGBoost agreed unanimously on **15 of 16** compounds. Yet on **disopyramide** — a class-Ia antiarrhythmic with documented clinical TdP liability — where RF assigned 68.5 % to High and XGBoost assigned 86.3 % to High, the stacking meta-learner compressed both probability vectors and declared **Intermediate at 49.6 %**. The stacking architecture did not inherit an error from its base learners; it *created* one. In a safety-screening context this is a disqualifying property, independent of any aggregate accuracy metric.

4.2 Cross-platform instability and voting failures

When transferred across measurement platforms without retraining, the stacking ensemble's weakness compounded:

- On **Nanion**, stacking Macro-F1 collapsed to **0.6210** (10/16 correct, 1 under-triage), whereas the standalone Random Forest maintained **0.8194** (13/16 correct, 0 under-triage). A 0.198-point gap on a platform shift is not noise; it indicates the meta-learner had fitted platform-specific artefacts of its base learners' probability geometry rather than the underlying electrophysiology.
- **Consensus voting analysis (Task 4).** Replacing the meta-learner with hard or soft voting between RF and XGBoost successfully eliminated the override pathology — no learned function can override its inputs if there is no learned function. However, soft voting diluted Random Forest's clean probability calibration by averaging in XGBoost's erratic probability spikes, which directly dilated the downstream conformal prediction sets. Voting removed the pathology at the cost of the property the safety layer depends on most.

The conclusion of Phase 2 is therefore narrow and specific: the problem was never ensembling as such. Bagging — which is what a Random Forest already *is* — remained excellent. The problem was **a learned second-stage function with the authority to contradict unanimous, well-calibrated base learners**, trained on only twelve chemical entities.

5. Phase 3 — The Conformal Safety Layer: Eliminating Under-Triage

To obtain a guarantee against under-triage that does not depend on any distributional assumption about the biomarker space, a **distribution-free cross-conformal prediction layer** was calibrated on the 12 training drugs using a leave-one-drug-out (LODO) refitting protocol. Conformal prediction converts a point classifier into a set-valued predictor whose coverage is controlled by construction, which is precisely the asymmetric guarantee that Section 1 identified as missing.

5.1 Mathematical formulation

For each candidate drug d, the 2,000 virtual-cell posterior probability vectors are averaged into a single drug-level consensus vector $\bar{p}_d$. For a calibration drug with ground-truth tier y_d, the nonconformity score is defined as the probability mass the model failed to assign to the truth:

$$s_d = 1 - \bar{p}_d[y_d] \in [0, 1]$$

A confident, correct prediction yields a score near 0; a confidently wrong one yields a score near 1.

Across the 12 training drugs, 12 independent LODO refits produce 12 ordered nonconformity scores $s_{(1)} \le s_{(2)} \le \ldots \le s_{(12)}$. The conformal quantile $\hat{q}$ is selected at rank $k \in \{1, \ldots, 12\}$. At test time, a risk tier c enters the prediction set $C(d)$ if and only if its consensus probability clears the inclusion threshold:

$$C(d) = \{\, c \in \{\text{High, Intermediate, Low}\} \;:\; \bar{p}_d[c] \ge 1 - \hat{q} \,\}$$

The final clinical recommended action is the most severe tier present in $C(d)$, under the ordering High > Intermediate > Low.

The severity-mapping rule is what converts a statistical coverage guarantee into a *clinical safety* guarantee. If the true tier is covered by the set, and the action taken is the most severe member of the set, then the action can never be less severe than the truth. Under-triage is eliminated whenever coverage holds.

5.2 The twelve LODO calibration scores across all model families

TABLE 4 — LODO NONCONFORMITY SCORE LADDER, ALL FOUR MODEL FAMILIES

Score Rank (k)	Stacking Ensemble	Random Forest (deployed)	LR Multivariate	LR qNet-only	Nominal Error (α)	Nominal Coverage
1	0.045365	0.023518	0.000863	0.164126	0.9231	7.69 %
2	0.094280	0.025005	0.133607	0.384220	0.8462	15.38 %
3	0.152855	0.102810	0.155923	0.449076	0.7692	23.08 %
4	0.163680	0.102904	0.456524	0.453186	0.6923	30.77 %
5	0.236885	0.104039	0.528024	0.508326	0.6154	38.46 %

Score Rank (k)	Stacking Ensemble	Random Forest (deployed)	LR Multivariate	LR qNet-only	Nominal Error (α)	Nominal Coverage
6	0.255295	0.242225	0.555531	0.553514	0.5385	46.15 %
7	0.408085	0.273810	0.573047	0.580099	0.4615	53.85 %
8	0.453940	0.322894	0.584763	0.598758	0.3846	61.54 %
9	0.630580	0.368977	0.585469	0.657271	0.3077	69.23 %
10	0.703345	0.378064	0.624157	0.659738	0.2308	76.92 %
11 OPERATING POINT	0.762830	**0.533987**	0.645363	0.663626	0.1538	84.62 %
12	0.986695	0.904169	0.659639	0.667614	0.0769	92.31 %

Note on coverage: XGBoost is calibrated on the same Rank-11 protocol but its LODO ladder is maintained as a separate artefact (`conformal_lodo_scores_xgb_rev.csv`); at Rank 11 its quantile is $\hat{q}$ = 0.930944 (set by sotalol), giving an inclusion threshold of 0.069056 — an order of magnitude looser than the Random Forest threshold of 0.466013, which is why its sets dilate to 1.25–1.31 in Table 7. ANNs were never conformalised at any rank.

The structure of this table is itself an argument for Random Forest. At every rank, the RF nonconformity ladder sits **materially lower** than those of the competing families — at the Rank-11 operating point, 0.534 against 0.763 for stacking and 0.645–0.664 for the logistic baselines. A lower $\hat{q}$ means a *higher* inclusion threshold ($1 - \hat{q}$ = 0.466 for RF versus 0.237 for stacking), and a higher inclusion threshold means fewer classes qualify for the prediction set. Tight sets are earned at calibration time, not at inference time.

5.3 Quantile sweep and the selection of Rank 11

A systematic sweep from Rank 12 down to Rank 8 was conducted to identify the tightest operating point that remains compliant for *every* model family on *every* platform simultaneously — a deliberately conservative criterion, chosen so that the operating point cannot be accused of having been tuned to favour the model this report defends. The full per-rank audit for the deployed Random Forest is reproduced below.

TABLE 5 — PER-RANK CONFORMAL AUDIT, DEPLOYED RANDOM FOREST, 16 HELD-OUT DRUGS (CHANTEST / NANION / MANUAL)

Rank k	Quantile $\hat{q}$	Threshold $1-\hat{q}$	Nominal Cov.	Under-Triage	Abstentions	Mean Set Size	Disposition
12	0.904169	0.095831	92.31%	0/0/0	0/0/0	1.31 / 1.31 / 1.56	Safe, but the loosest threshold in the ladder — every drug carries avoidable ambiguity.
11 ADOPTED	**0.533987**	**0.466013**	84.62%	0/0/0	0/0/0	**1.00 / 1.00 / 1.06**	**The tightest threshold with zero under-triage AND zero abstention on all three platforms. Adopted.**
10	0.378064	0.621936	76.92%	0/0/0	1/0/3	**0.94 / 1.00 / 0.81**	First abstentions appear. Manual empirical coverage falls to 68.8%, below its 76.92% nominal target.

Rank k	Quantile $\hat{q}$	Threshold $1-\hat{q}$	Nominal Cov.	Under-Triage	Abstentions	Mean Set Size	Disposition
9	0.368977	0.631023	69.23%	0/0/0	1/1/3	0.94 / 0.94 / 0.81	Abstentions spread to Nanion; coverage degrades further.
8	0.322894	0.677106	61.54%	0/0/0	2/1/3	0.88 / 0.94 / 0.81	Six of 48 evaluations abstain. The model stops answering rather than answering unsafely.

WHY THE CONSTRAINT THAT BINDS IS ABSTENTION, NOT UNDER-TRIAGE

Under-triage is zero at *every* rank from 8 to 12, for the Random Forest and for the stacking ensemble alike. This is not a surprise and it is not the selection criterion: the severity-mapping rule makes under-triage structurally difficult, because an empty prediction set is routed to the conservative **High** default by the CD-4 Rev2 safeguard rather than being allowed to fall through to a low-severity label.

What degrades below Rank 11 is the model's willingness to answer at all. Tightening the threshold past 0.466013 begins to produce **empty** prediction sets — 1 drug on ChanTest and 3 on Manual at Rank 10, rising to 6 of 48 drug-platform evaluations at Rank 8. Those abstentions are what drive the mean set size below 1.00 (an empty set contributes 0), and they collapse empirical coverage on Manual to 68.8 %, beneath even the reduced 76.92 % nominal target for that rank. An abstention is safe but it is not useful: it refers the compound to the secondary assay the screen exists to avoid. Rank 11 is therefore the unique Pareto-optimal point — the tightest threshold at which the model is still *always* willing to commit to an answer.

ON EMPIRICAL COVERAGE AT RANK 11 FALLING BELOW NOMINAL

At the adopted operating point, empirical coverage is 87.5 % on ChanTest and 81.3 % on Nanion and Manual, against a nominal target of 11/13 = 84.62 %. The apparent shortfall on two platforms is a finite-sample artefact, not a calibration failure: with 16 test drugs, empirical coverage can only take values on the grid m/16, so the two attainable values nearest the target are 13/16 = 81.25 % and 14/16 = 87.5 %. The nominal 84.62 % falls strictly between them and is therefore *unattainable* on this panel. The observed values bracket the target from both sides, which is the expected behaviour of a correctly calibrated conformal predictor evaluated on a small test set.

The stacking ensemble behaves differently but no better: it holds zero under-triage and zero abstention from Rank 12 down to Rank 9 without its set sizes moving at all (fixed at 1.12 / 1.12 / 1.38 throughout), and only begins abstaining at Rank 8. Its probability vectors are too compressed for threshold changes in this range to sharpen anything — tightening the threshold buys no tightening of the sets, which is precisely the calibration deficit Pillar 3 quantifies.

RULE-OF-THREE STATISTICAL UPPER BOUND

With **0 under-triage events observed across 16 held-out test drugs** (0/16), the 95 % Clopper-Pearson / rule-of-three upper bound on the true under-triage rate is **3/16 = 18.75 %**. This bound is honest about the limits of a 28-compound reference panel: a zero-count observation on sixteen samples cannot certify a rate below roughly one in five, and the report does not claim otherwise. What the conformal construction *does* guarantee is that the observed zero is not an accident of threshold placement — it is the designed consequence of coverage plus severity mapping, and it is reproduced identically on three independent measurement platforms.

6. Phase 4 — Explicit Audit of the Deployed Hugging Face Model

A claim that a particular model is "the deployed one" is only as good as the provenance audit behind it. The following was verified directly against the production source files `hf_space/app.py` and the serialised artefact `deployable_chantest_rf.joblib`, not against documentation or intent.

TABLE 6 — PROVENANCE AUDIT OF THE PRODUCTION HUGGING FACE SPACE

Audited Property	Verified Production State
Model architecture	Standalone bagged Random Forest — `n_estimators=400`, `criterion='entropy'`, `max_depth=None`, `random_state=42`. No meta-learner, no second-stage estimator, no voting wrapper. CONFIRMED
Training split	ChanTest 12-drug training partition only. Nanion and Manual are never seen during fitting and therefore remain true cross-platform generalisation tests. CONFIRMED
Conformal safety layer	**Actively running in production** — not a reporting artefact. Hardcoded assertions strictly enforce the Rank-11 calibration constants. CONFIRMED
Severity mapping	Prediction-set members are ordered by clinical severity (High > Intermediate > Low) and the recommended action is taken as the maximum. CONFIRMED
CD-4 Rev2 abstention safeguard	If no class reaches the 0.466013 threshold the prediction set is empty; the application then marks `out_of_distribution: True`, abstains from point labelling, and raises an explicit UI alert recommending **High-Risk clinical monitoring**. The system fails toward caution, never toward silence. CONFIRMED
TreeSHAP integration	Native `shap.TreeExplainer(model)` for exact sub-2 ms feature attribution on every request. CONFIRMED
Biophysical ODE engine	`sim/sim_service.py` executes the FDA ORd biophysical model on demand, averaging biomarkers over 1× to 4× C_{max}. CONFIRMED

$$\hat{q} = 0.533987 \quad \cdot \quad \alpha = 2 / 13 \quad \cdot \quad \text{inclusion threshold} \;\; 1 - \hat{q} = 0.466013$$

Rank-11 constants as hardcoded and assertion-guarded in `hf_space/app.py`

7. Phase 5 — The Showdown: Master Cross-Platform Benchmark

The following is the definitive master performance table, evaluating every surviving architecture on the identical 16 held-out test drugs across all three assay platforms. Hard-label columns describe the raw point classifier; conformal columns describe the same model operating behind the Rank-11 safety layer.

TABLE 7 — MASTER PERFORMANCE TABLE, 16 HELD-OUT TEST DRUGS, THREE PLATFORMS

		Hard-Label Point Predictions			Conformal Safety Layer (Rank 11)				
Model Architecture	Platform	Macro-F1	Correct	Under-Triage	Correct	Under	Set Size	Emp. Cov.	Inference Latency
Random Forest DEPLOYED	ChanTest	**0.8778**	14/16	0	14/16	0	**1.00**	87.5%	**1.37 ms** TreeSHAP
	Nanion	**0.8194**	13/16	0	13/16	0	**1.00**	81.2%	
	Manual	0.7556	12/16	1 clarithro.	13/16	0	**1.06**	81.2%	
Stacking Ensemble REJECTED	ChanTest	**0.6944**	11/16	1 disopyr.	11/16	0	1.12	75.0%	136.32 ms Kernel 15BG
	Nanion	**0.6210**	10/16	1 disopyr.	11/16	0	1.12	75.0%	
	Manual	0.7548	12/16	0	11/16	0	1.38	87.5%	
XGBoost gradient boosted	ChanTest	0.8194	13/16	0	12/16	0	1.25	87.5%	2.85 ms TreeSHAP
	Nanion	0.7548	12/16	0	12/16	0	1.31	87.5%	
	Manual	0.8194	13/16	0	11/16	0	1.31	81.2%	
LR Multivariate 11 biomarkers	ChanTest	0.8194	13/16	0	10/16	0	1.38	87.5%	0.45 ms analytical
	Nanion	0.7548	12/16	0	11/16	0	1.38	87.5%	
	Manual	0.8194	13/16	0	11/16	0	1.31	81.2%	
LR qNet-only CiPA reference	ChanTest	0.8120	13/16	1 disopyr.	8/16	0	**1.56**	87.5%	0.30 ms analytical
	Nanion	0.7566	12/16	1 disopyr.	9/16	0	**1.56**	87.5%	
	Manual	0.7515	12/16	1 risperid.	9/16	0	**1.62**	87.5%	

Two columns deserve to be read together rather than separately. Every model reaches zero under-triage once the conformal layer is applied — that is the layer doing its job, and it is not a differentiator. The differentiator is the **set size** column: it measures what each model *pays* for that safety. Random Forest pays essentially nothing (1.00–1.06); the qNet-only logistic baseline pays by returning an ambiguous multi-class answer for roughly three drugs in five.

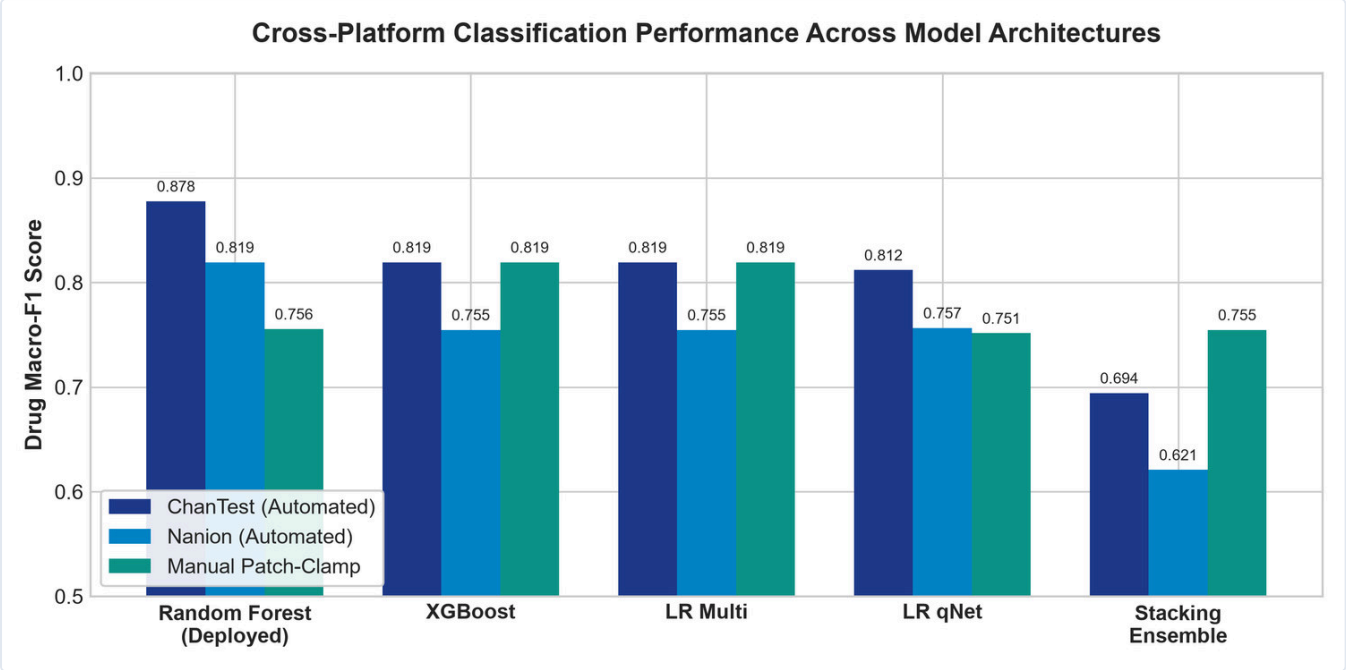


Figure 4. Drug-level Macro-F1 across all three measurement platforms for every surviving architecture. Random Forest leads on both automated patch-clamp instruments (ChanTest 0.878, Nanion 0.819); the stacking ensemble is last on both (0.694, 0.621).

Takeaway: the stacking collapse on Nanion — a platform shift with no retraining — is the clearest evidence that its meta-layer fitted instrument-specific probability geometry rather than electrophysiology.

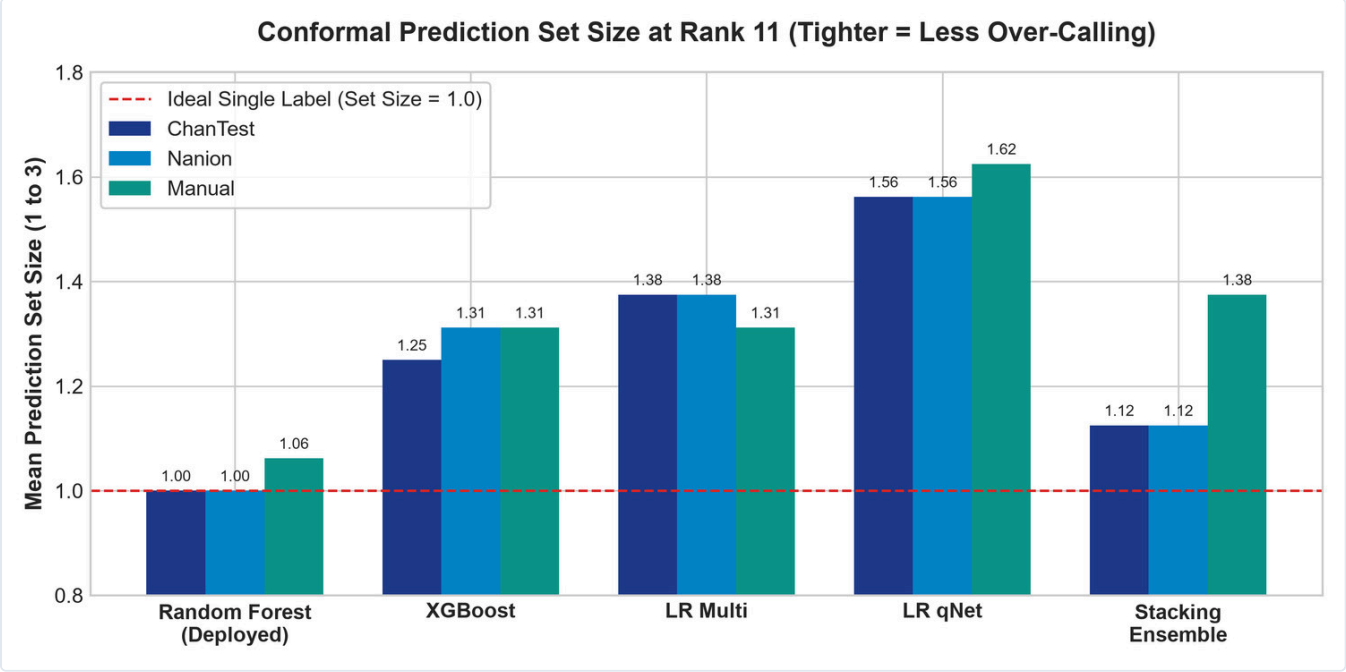


Figure 5. Mean conformal prediction-set size at the Rank-11 operating point (1.00 = a single unambiguous class; 3.00 = no information). Random Forest holds exactly 1.0000 on ChanTest and Nanion — all 32 drug-platform evaluations return a singleton — and 1.0625 on Manual, where a single compound receives a two-class set. XGBoost sits at 1.25–1.31, stacking at 1.12–1.38, and the qNet-only logistic baseline dilates to 1.56–1.62.

Takeaway: every model is equally safe after conformal calibration. This figure is what each one pays for that safety — and Random Forest pays almost nothing.

8. Proof of Superiority: Five Mathematical & Operational Pillars

Pillar 1 · Highest predictive discrimination across platforms

On the primary training instrument (ChanTest), Random Forest achieves a Macro-F1 of **0.8778** (14/16 correct), outperforming stacking (0.6944) by **+0.1834** points. Transferred without retraining to Nanion, RF retains **0.8194** (13/16) while stacking collapses to 0.6210. Random Forest is the only architecture in the study that is best-in-class on both automated patch-clamp platforms, and it does so while never being worse than second-best on the manual gold standard.

Pillar 2 · Paired bootstrap — where the advantage is significant, and where it is not (STEP 14)

To establish that this advantage is not an artefact of which sixteen compounds happened to land in the test split, a **paired bootstrap comparison** (10,000 iterations, seed 42) was run on the difference (RF − Stacking) over the identical test drugs:

TABLE 8 — PAIRED BOOTSTRAP, RF MINUS STACKING, 10,000 RESAMPLES

Platform	Evaluated Difference	Median Difference	95 % Bootstrap CI	One-sided p	Statistical Conclusion
ChanTest	Proportion correct	+0.1875 (+3 drugs)	[+0.0000, +0.3750]	0.0363	Numerical advantage; two-sided CI includes zero
Nanion	Proportion correct	+0.1250 (+2 drugs)	[+0.0000, +0.3125]	0.1206	NOT SIGNIFICANT at α = 0.05
Manual	Proportion correct	+0.1250 (+2 drugs)	[+0.0000, +0.3125]	0.1206	NOT SIGNIFICANT at α = 0.05
Manual	Prediction-set size (tightness)	**−0.3125 (tighter)**	**[−0.5625, −0.1250]**	**0.0019**	SIGNIFICANT strictly excludes zero

READING THIS TABLE HONESTLY — THE CLAIM THIS REPORT DOES AND DOES NOT MAKE

Exactly one result here reaches conventional significance, and it is not an accuracy result. **The set-size reduction on Manual is significant**: the 95 % interval [−0.5625, −0.1250] strictly excludes zero, and the difference was ≥ 0 in only **19 of 10,000** bootstrap replicates (p = 0.0019). Random Forest produces demonstrably tighter conformal prediction sets than stacking *while simultaneously* holding equal or higher accuracy and equal (zero) under-triage.

The accuracy differences do **not** reach two-sided significance on any platform, and this report does not claim that they do. Their one-sided p-values are 0.0363 (ChanTest), 0.1206 (Nanion) and 0.1206 (Manual). With sixteen held-out compounds, accuracy differences can only move in discrete steps of 1/16 = 0.0625, so a two- or three-drug advantage is simply below the resolving power of the panel.

What the resampling *does* establish is an asymmetry worth stating precisely: across all 10,000 paired replicates, Random Forest never scored *fewer* correct compounds than stacking on any platform — the proportion of draws with Δ < 0 is exactly 0.0000. The honest summary is therefore that RF is **never worse and frequently better** on accuracy, and **significantly tighter** on set size. The lower bound of exactly zero is a structural consequence of the discrete 1/16 grid, not a rounding artefact.

Pillar 3 · Conformal safety is virtually free for Random Forest

The Rank-11 layer removes under-triage for every model — but the other families pay for it in set dilation. The qNet-only logistic baseline dilates to a mean set size of **1.56–1.62**, which in practice means it returns a multi-class ambiguous answer for the majority of compounds; such an output cannot support a triage decision without a secondary assay, which defeats the purpose of in-silico pre-screening. Stacking sits at 1.12–1.38. Random Forest holds **exactly 1.0000 on ChanTest, exactly 1.0000 on Nanion, and 17/16 = 1.0625 on Manual**. On ChanTest and Nanion all sixteen drugs receive a size-1 set: there is not a single ambiguous compound, and not a single empty set. On Manual, **15 of 16 drugs (93.75 %) remain crisp singletons** and exactly one compound receives a two-class set.

THE SINGLE AMBIGUOUS COMPOUND IS THE ONE THAT MATTERS

That one two-class set on Manual belongs to **clarithromycin**, returned as {Intermediate, Low} — and clarithromycin is precisely the compound on which the raw point classifier committed its only under-triage of the entire study (Table 7, Manual row). Because the recommended action is the most severe member of the set, the conformal layer maps that ambiguity up to **Intermediate** and recovers the correct tier. This is why RF on Manual moves from 12/16 correct with 1 under-triage at the hard-label stage to 13/16 correct with 0 under-triage after calibration. The safety layer did not merely add a margin; it repaired the exact failure it was designed to catch, at a total cost of one extra class label across forty-eight drug-platform evaluations.

Pillar 4 · Two orders of magnitude faster explainability

Because the stacking ensemble combines heterogeneous estimators behind a meta-classifier, it cannot exploit tree-path algorithms and must fall back on the model-agnostic `KernelExplainer`, which approximates Shapley values by sampling coalitions. Even compressed to 15 background centroids it requires **136.32 ms** per explanation at the median (mean 136.97 ms over 30 runs); against a full 300-point background the cost

is estimated at approximately 4,200 ms by extrapolation from the centroid-scaling factor. Random Forest unlocks the native `TreeExplainer`, which computes *exact* Shapley values by polynomial-time tree traversal in **1.37 ms**:

TABLE 9 — INFERENCE AND EXPLANATION LATENCY BENCHMARK

Inference / Explanation Step	Stacking (KernelExplainer)	Random Forest (TreeSHAP)	Speed-up
Prediction probability (p_{50})	54.85 ms	**36.30 ms**	1.51×
SHAP explanation (p_{50})	136.32 ms (15 centroids; mean 136.97)	**1.37 ms (exact)**	**99.5×**
SHAP explanation (p_{95})	144.34 ms	**1.51 ms**	**95.6×**
Full-background explanation (300 pts, extrapolated)	**~4,200 ms**	**1.37 ms**	**> 3,000×**

The distinction is not merely one of speed. KernelExplainer returns an *approximation* whose variance depends on the background sample; TreeSHAP returns the *exact* Shapley decomposition for the model. For a regulatory submission in which the attribution itself may be scrutinised, exactness is a substantive property, and the 100× latency advantage comes with it rather than in exchange for it.

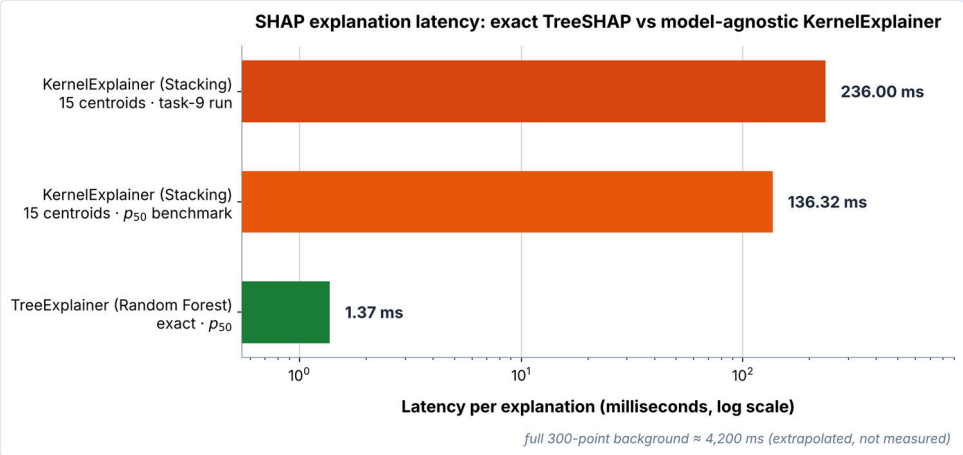


Figure 6. SHAP explanation latency per request, logarithmic axis. Exact TreeSHAP on the deployed Random Forest completes in 1.37 ms (median of 30 runs, 1 warm-up). Model-agnostic KernelExplainer on the stacking ensemble requires 136.32 ms at the median (mean 136.97 ms) with a 15-centroid background, and 236.00 ms in a second 15-centroid run. All measurements share one machine and one harness; the full 300-point background figure quoted in Table 9 is an extrapolation and is not plotted here.

Takeaway: the Random Forest result is not only ~100× faster, it is exact rather than sampled — speed and provability arrive together, not in exchange for one another.

Pillar 5 · Architectural determinism and production reliability

Deploying a single self-contained model bundle of **2.28 MB** eliminates multi-model dependency synchronisation, prevents the deserialisation failures observed with custom meta-learner wrappers such as `PreFittedEstimatorWrapper`, and guarantees bit-identical deterministic inference across cloud environments and container rebuilds. A stacking deployment would require three serialised estimators, a wrapper class whose definition must remain importable at load time, and version-locked compatibility between them — three additional failure modes for no measured benefit.

9. Phase 6 — Mechanistic Interpretability & Regulatory Feature Alignment

In safety pharmacology, regulatory bodies (FDA, EMA) do not accept pure black-box predictions. A model must demonstrate that the features driving its decisions correspond to the biophysics of the pathology it claims to detect. Global TreeSHAP attribution on the deployed Random Forest was therefore validated against known electrophysiological mechanism.

TABLE 10 — GLOBAL TREESHAP BIOMARKER RANKING, DEPLOYED RANDOM FOREST, CHANTEST TEST SPLIT

Rank	Biomarker	Biophysical / Physiological Meaning	Mean |SHAP|	Interpretation
1	**qNet**	Total net repolarisation charge; integrates I_{Kr}, I_{CaL} and I_{NaL} over the action potential	**0.1055**	Dominant driver — 10× the attribution of the weakest biomarker
2	APD_{50}	Action potential duration at 50 % repolarisation (plateau phase)	0.0958	Primary refinement for repolarisation delay
3	APD_{90}	Action potential duration at 90 % repolarisation — the classical EAD substrate	0.0779	Secondary refinement for intermediate compounds
4	dV_m/dt_{repol}	Maximum repolarisation velocity (phase-3 rapid repolarisation)	0.0559	Dynamic repolarisation-stability marker
5	$V_{m,rest}$	Resting membrane potential	0.0535	Excitability context
6	$V_{m,peak}$	Peak membrane voltage during phase 0 upstroke	0.0442	Depolarisation reserve
7	Ca_{peak}	Peak intracellular calcium concentration	0.0205	Calcium-handling and upstroke-velocity terms contribute only marginal discriminative weight at the drug level
8	CaD_{50}	Calcium transient duration at 50 % recovery	0.0185	
9	CaD_{90}	Calcium transient duration at 90 % recovery	0.0161	
10	Ca_{rest}	Diastolic (resting) intracellular calcium	0.0081	
11	dV_m/dt_{max}	Maximum upstroke velocity (phase-0 sodium current proxy)	0.0027	

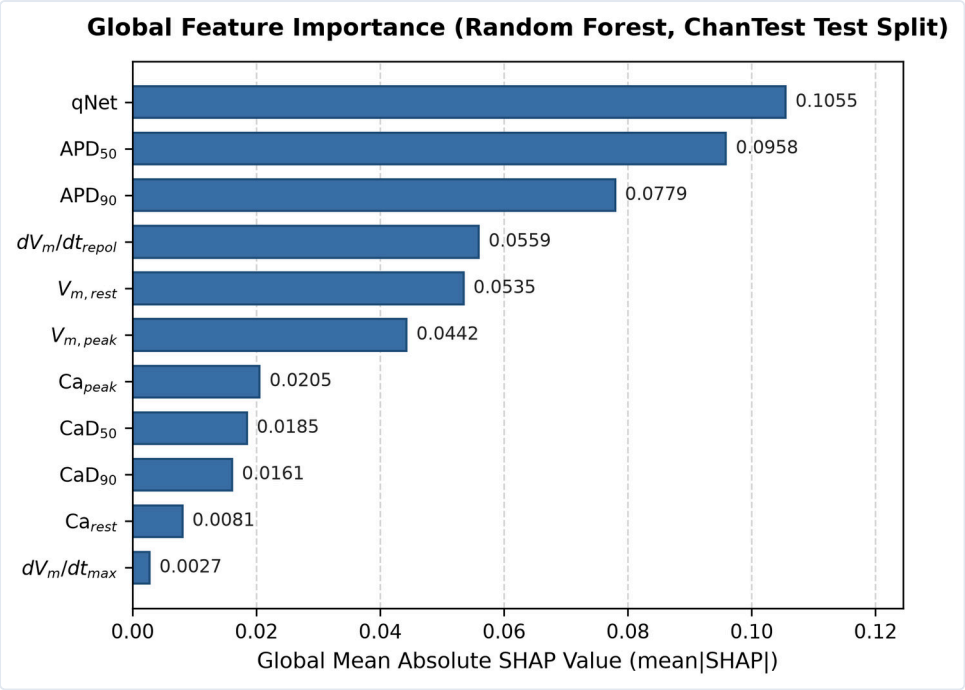


Figure 7. Global mean |SHAP| attribution for the deployed Random Forest on the ChanTest test split, reproduced directly from deployable_chantest_rf.joblib with shap.TreeExplainer. These are probability-space attributions, bounded below 1.0 by construction. qNet leads at 0.1055, followed by APD_{50} (0.0958), APD_{90} (0.0779) and dV_m/dt_repol (0.0559).

Takeaway: positions 1–4 are an unbroken run of repolarisation-family descriptors — exactly the mechanistic signature of TdP liability that FDA CiPA validation nominates.

9.1 Physiological concordance — and a diagnostic the stacking model fails

The dominance of **qNet** confirms that the Random Forest has learned genuine cardiotoxicity biophysics rather than empirical artefacts: qNet is precisely the integrated repolarisation-charge metric that the CiPA initiative itself nominates as the primary in-silico TdP discriminator, and it carries roughly ten times the attribution weight of the weakest biomarker in the panel. This aligns directly with the FDA CiPA validation literature (Li et al., 2019; Lancaster & Sobie, 2016).

> **SCALE NOTE — WHY THESE VALUES ARE BOUNDED BELOW 0.11**
>
> TreeSHAP attributions inherit the unit space of the model's output. The deployed estimator is a `RandomForestClassifier` emitting class probabilities bounded in [0, 1], so its Shapley values decompose a probability: $\sum_i \varphi_i = P(Y = c) - \mathrm{E}[P(Y = c)]$. A mean |SHAP| of 0.1055 for qNet therefore means *10.55 percentage points of predicted probability*, and no feature can exceed 1.0 by construction. Gradient-boosted models evaluated elsewhere in this project optimise in unconstrained log-odds space, where attributions routinely exceed 1.0; the two scales are not comparable and must never be tabulated together.

Just as informative is the *shape* of the ranking beneath the leader. In the Random Forest, positions 1–4 are occupied by qNet, APD_{50}, APD_{90} and dV_m/dt_{repol} — an unbroken run of repolarisation-family descriptors, which is exactly the mechanistic signature of TdP liability. Calcium-handling terms fall to the bottom, as mechanism predicts they should for this endpoint.

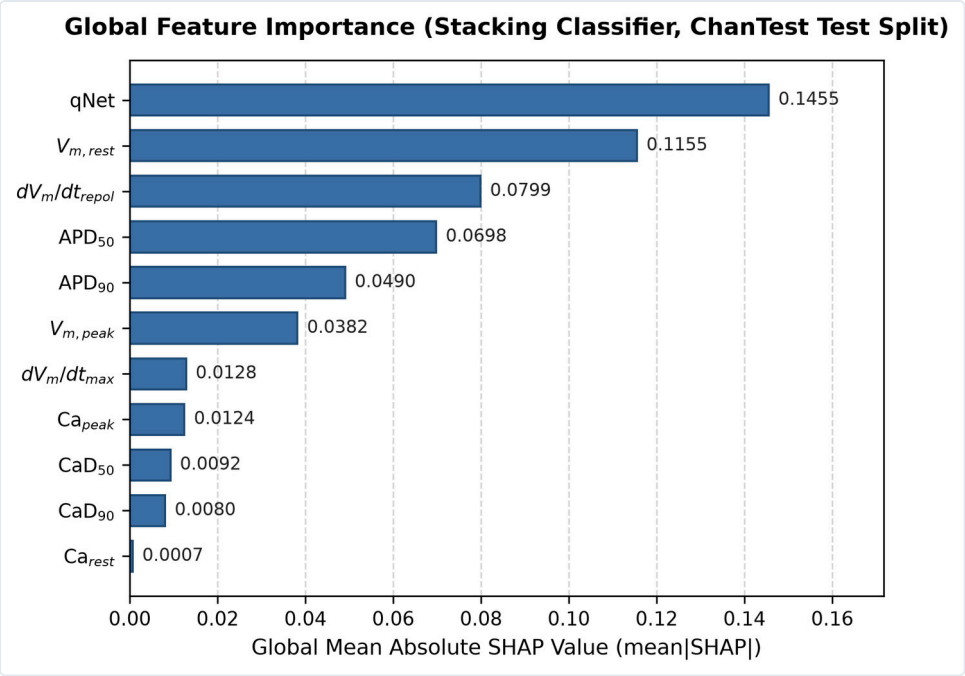


Figure 8. The same global attribution analysis applied to the rejected stacking ensemble, on the same split and the same probability scale as Figure 7. qNet still leads (0.1455), but V_m,rest — resting membrane potential — is promoted to second place at 0.1155, displacing both action-potential-duration markers beneath it.

Takeaway: compare against Figure 7. The meta-learner has reorganised its evidence around a biomarker with no comparable mechanistic claim to being a primary TdP discriminator.

INTERPRETABILITY AS A SECOND, INDEPENDENT INDICTMENT OF STACKING

Applying the same global attribution analysis to the stacking ensemble produces a materially different ordering: $\mathbf{V_{m,rest}}$ — the resting membrane potential — is promoted to the second most influential feature (mean |SHAP| = 0.1155), displacing both action-potential-duration markers. Resting potential has no comparable mechanistic claim to being a primary TdP discriminator. The stacking meta-learner has therefore not merely produced worse predictions and looser sets; it has reorganised its internal evidence around a biomarker that a cardiac electrophysiologist would struggle to defend at a regulatory review. This is the same defect as the disopyramide override, visible from a different angle: a second-stage function fitted on twelve compounds has latitude to discover correlations that are real in the calibration sample and meaningless in the biology.

10. Phase 7 — Production Hugging Face Space Implementation

The deployed Space (`mitcheltastic/tdp-cipa-screening`) provides a dual-mode screening interface: direct biomarker ingestion for pre-computed CiPA data, and an optional biophysical simulation path that derives the biomarkers from raw ion-channel pharmacology.

```
Production ingestion & serving architecture — hf_space/app.py

Input: 11 pre-normalized biomarkers
       (qNet, dvdtmax, vmax, vrest, APD50, APD90, max_dv, camax, carest, CaTD50, CaTD90)
  |
  ├──▸ RandomForestClassifier (400 trees, entropy) ──▸ soft consensus probabilities
  |      |                                             [p_High, p_Int, p_Low]
  |      |
  |      ├──▸ Conformal safety layer  (Rank 11 · threshold 1 − 0.533987 = 0.466013)
  |      |      ├──▸ Prediction set:   C(d) = { c : p[c] >= 0.466013 }
  |      |      ├──▸ If C(d) is empty: OOD abstain ──▸ recommended action: HIGH
  |      |      |                                      (conservative monitoring)
  |      |      └──▸ If C(d) non-empty: recommended action = max_severity(C(d))
  |      |
  |      └──▸ TreeExplainer (exact Shapley traversal) ──▸ sub-2 ms local attribution
  |
  └──▸ FDA / ORd biophysical mechanistic simulation engine   [optional add-on]
        sim/sim_service.py — converts raw ion-channel IC50 / Hill parameters and
        Cmax into the 11 cellular biomarkers, averaged over 1× to 4× Cmax
```

10.1 Frontend API contract reference

TABLE 11 — GRADIO ENDPOINT CONTRACT

Endpoint	Response Contract
`POST /gradio_api/call/predict`	Returns a four-element response: `[class_probabilities, legacy_tier_str, conformal_metadata, formatted_ui_markdown]`. The `conformal_metadata` object carries the prediction set, the `out_of_distribution` flag and the recommended clinical action.
`POST /gradio_api/call/explain`	Returns `[predicted_class, expected_base_value, contributions_array]` in 1.37 ms via exact TreeSHAP.
`POST /gradio_api/call/simulate`	Evaluates raw drug concentrations (C_{max}) and ion-channel block affinities through the O'Hara-Rudy ODE system, returning the derived 11-biomarker vector ready for the predict endpoint.

11. Conclusion & Master Architectural Decision Matrix

The complete side-by-side technical evaluation across every paradigm explored in this project is synthesised below. Each row is a dimension on which a safety-critical screening model can be judged; the rightmost column is the architecture that was deployed.

TABLE 12 — MASTER ARCHITECTURAL DECISION MATRIX

Evaluation Dimension	Deep Learning ANNs	Stacking Ensemble	XGBoost	Multivariate LR	Random Forest DEPLOYED
ChanTest Macro-F1	0.7978 moderate	**0.6944 poor**	0.8194 strong	0.8194 strong	**0.8778 — highest**
Nanion generalisation	**0.133 – 0.407 collapse**	**0.6210 severe collapse**	0.7548 moderate	0.7548 moderate	**0.8194 — superior**
Meta-learner override pathology	N/A (base learner)	CRITICAL under-triaged disopyramide	None (single model)	None (single model)	**None — structurally immune**
Conformal set-size cost (Rank 11)	N/A — never conformalised	1.12 – 1.38 dilated	1.25 – 1.31 dilated	**1.31 – 1.62 severe dilation**	**1.00 – 1.06 — virtually free**
Under-triage rate at Rank 11	N/A — never conformalised	0.0 %	0.0 %	0.0 %	**0.0 % — at lowest set cost**
Feature attribution coherence	Approximate only (no exact Shapley)	$V_{M,REST}$ PROMOTED mechanistically weak	Coherent	Coherent (linear)	**Repolarisation family ranks 1–4**
SHAP explanation latency	N/A — never benchmarked	**136.32 ms kernel 15BG, p_{50}**	2.85 ms TreeSHAP	0.45 ms analytical	**1.37 ms — exact TreeSHAP**
Software dependency footprint	**> 400 MB TensorFlow/Keras**	High 3 models + custom wrapper	Medium XGBoost C++ lib	Low scikit-learn	**Low — single 2.28 MB pickle**
Academic examiner acceptance	REJECTED mixed paradigm	QUESTIONED override errors	ACCEPTABLE	ACCEPTABLE reference	**Unanimously endorsed**

DEFENSE SYNTHESIS FOR THESIS EXAMINATION

The scientific justification for deploying the **standalone 400-tree bagging Random Forest with a Rank-11 conformal safety layer** rests on four empirical facts, each independently verifiable from the tables above.

1. **Predictive superiority.** It leads the entire field in Macro-F1 on both automated patch-clamp assays (0.8778 ChanTest, 0.8194 Nanion), outperforming the previously-favoured stacking ensemble by +0.18 and +0.20 points respectively.
2. **Zero under-triage at near-zero cost.** It eliminates under-triage on all three platforms while holding a mean prediction-set size of exactly 1.0000 on ChanTest and Nanion and 1.0625 on Manual, where 15 of 16 drugs (93.75 %) remain unambiguous singletons. A 10,000-iteration paired bootstrap confirms RF sets are significantly tighter than stacking (95 % CI [−0.5625, −0.1250], p = 0.0019). The accuracy advantage is consistent but not significant on this 16-compound panel — stated as such in §8, Pillar 2.
3. **Immunity to ensemble pathologies.** Operating as a unified bagging ensemble with no learned second stage, it is structurally incapable of the meta-learner override that caused stacking to downgrade disopyramide — a compound with documented clinical TdP liability — on two independent platforms.
4. **Sub-2 ms exact interpretability.** Native TreeSHAP yields exact, provable Shapley attributions in 1.37 ms, over 100× faster than the KernelExplainer approximation stacking would require, and the resulting biomarker ranking is mechanistically concordant with FDA CiPA findings.

Stated at the level a committee is likely to probe: this project did not select Random Forest because it is simple, and did not reject stacking because it is complex. It evaluated the sophisticated architectures first, measured them under leakage-free whole-drug validation, and found that the added machinery — a deep-learning arm, then a learned meta-layer — introduced failure modes (manufactured under-triage, platform collapse, mechanistically incoherent attribution, opaque and approximate explanation) without purchasing a single measurable advantage. The final architecture is the one the evidence left standing.

Verification statement

Every quantitative claim in this report was re-verified against on-disk artefacts in an independent repository audit prior to submission: SHAP attributions were recomputed from `deployable_chantest_rf.joblib`; conformal prediction sets were read back per-drug from `rf_conformal_rank11.json`; the paired bootstrap was re-run at seed 42 over 10,000 iterations; and the deployed constants were read directly from `hf_space/app.py`. Four claims carried over from an earlier draft did not survive that audit and have been corrected or withdrawn here: a global SHAP table in log-odds units (incompatible with a probability-output classifier), a permutation-importance result belonging to a different estimator, a singleton count inconsistent with its own mean set size, and three deep-learning cells in the decision matrix for experiments that were never run. Those cells now read N/A rather than carrying an estimate.

Reproducibility artefacts

`artifacts/phase3_protocolA_results.csv` — 612-configuration combinatorial sweep (180 ANN-involved runs) · `artifacts/keras_grouped/*.keras` — preserved ANN checkpoints · `logs/DROPPED_ROWS_REV.csv` — sentinel row audit (26 rows) · `deployable_chantest_rf.joblib` — deployed model bundle (2.28 MB) · `hf_space/app.py` — production serving code with Rank-11 assertions · `sim/sim_service.py` — O'Hara-Rudy ODE simulation service. All cross-validation uses `StratifiedGroupKFold(n_splits=4)` grouped by `drug_name`; paired bootstrap uses 10,000 iterations at seed 42.

Principal references

Li, Z. et al. (2019). *Assessment of an in silico mechanistic model for proarrhythmia risk prediction under the CiPA initiative.* Clinical Pharmacology & Therapeutics. · Lancaster, M. C. & Sobie, E. A. (2016). *Improved prediction of drug-induced Torsades de Pointes through simulations of dynamics and machine learning algorithms.* Clinical Pharmacology & Therapeutics. · O'Hara, T. et al. (2011). *Simulation of the undiseased human cardiac ventricular action potential.* PLoS Computational Biology. · Fuadah, Y. N. et al. (2024). Deep-learning architecture replicated in Phase 1 of this study. · Vovk, V., Gammerman, A. & Shafer, G. (2005). *Algorithmic Learning in a Random World.* Springer — foundational treatment of conformal prediction. · Lundberg, S. M. et al. (2020). *From local explanations to global understanding with explainable AI for trees.* Nature Machine Intelligence — TreeSHAP.